

BOX 2026 – Homework 2

September 9, 2026

Instructions

1. Download all the files from the homework2 directory in

https://drive.google.com/drive/folders/1jna1Swlrn_AKLHUGjrfL5Bhbt4EG90Nu

2. For each FASTA file, compute and print the number of sequences and the cumulative sequence length. What can you tell about the content of these files from these numbers?
3. Use NCBI BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) to find out which species these sequence come from and possibly more.
4. For $k = 20$, compute the set of canonical k -mers of every FASTA file, and then the pairwise Jaccard indexes

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

What does this tell you about the genomes and how similar they are?

5. To save memory, now store the k -mers as numbers, not as strings. Encode each nucleotide using two bits.
6. Implement canonical k -mer counting, compute the k -mer abundance histogram for one of the files, and plot it using your preferred method.

Notes

- From now on, always use the attached generator function `readfq` to parse FASTA and FASTQ files.
- If you have not heard the term before, find out what a “canonical k -mer” is on the Internet.
- Don’t forget to convert all letters consistently to either lower case or upper case
- Some letters can encode ambiguities (e.g., Y = C or T). Decide explicitly what your program should do when it encounters such letters.